

Bethany Spear

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PROFILE

I am a DPhil Candidate at the John Adams Institute for Accelerator Research based at the University of Oxford. I am broadly interested in accelerator diagnostics and beam instrumentation, especially in response to novel beam-line designs and challenges arising from new acceleration techniques. My current research focuses on producing different high frequency electron beam position monitors for the Advanced Wakefield Experiment (AWAKE) at CERN.

During my studies I have had numerous opportunities to present my research to a wide audience of academics, students and the general public in the form of posters, presentations and papers, developing my strong science communication skills.

EDUCATION

DPhil Accelerator Physics 2021 - May 2025 (*expected*)

University of Oxford, UK

Thesis: GHz Frequency Beam Position Monitor Solutions for the Advanced Wakefield Experiment

Supervisors: Prof. Phil Burrows (Oxford) and Manfred Wendt (CERN)

MSc Particle Physics 2017 - 2021

University of Liverpool, UK

Thesis: Optimization of a supersonic gas jet-based beam profile monitor for the LHC

Supervisors: Prof. Carsten Welsch and Dr. Hao Zhang

OUTREACH AND COMMUNITY ENGAGEMENT

Summer Camp Counsellor 2019, 2020

I worked in China as a camp leader for international school children aged 7–14. I led the 'Adventure' team, planning activities like climbing, woodworking, and rope courses, while also providing pastoral care and conflict mediation. Working with a diverse staff, I fostered a supportive environment and navigated cultural and language barriers with patience and empathy.

Pint Of Science Public Talks 2023, 2024

Presentation of the workings of the CERN accelerator complex to the general public as part of the *Pint of Science Festival*. In 2024, I organised the event in my College Bar, recruiting researchers from a variety of fields to present their research alongside my own talk.

CERN Tour Guide 2022-2023

I worked as a CERN trained tour guide for organised tours of the CERN visit points (ATLAS Cavern, LEIR+ LINAC2, Cyclosynchrotron and CLEAR), for members of the general public and school groups.

Department Outreach 2022

Funky Drums: Combining physics and music to engage young people with accelerator science.

An exercise for 11-15 year old students using custom made Bodhran Drums to demonstrate RF Cavity mechanisms.

The Accelerate! Show

An interactive demonstration lecture about accelerator science.

WORK EXPERIENCE AND LEADERSHIP ROLES

Stock Manager, Wolfson College Bar

2022, 2023-2025

While based in Oxford I took an executive Committee role with my College bar, responsible for stocking, pricing, and associated day-to-day operations of the bar. I worked closely with our suppliers and pushed for new relationships with stockists, as well as maintaining good relations with the College administration. This role required high levels of organisation, commitment and ability to work well in a team to produce a termly social calendar.

Research and Development Internship, QUASAR group

June – September, 2020

The Cockcroft Institute of Accelerator Science and Technology

Worked with researchers within the QUASAR team developing the novel Supersonic Gas Beam Profile Monitor for the Hi-Luminosity LHC.

Created simulations using MATLAB to better understand gas propagation through the device.

Produced outreach materials using the animation software Blender to contribute to the institute's outreach programme.

IT SKILLS

Computing: I am proficient in Python and MATLAB, and compiled basic Bash scripts. I have experience in Windows and Linux environments.

Software: Daily use of Microsoft office (Excel, PowerPoint, Office) and LaTeX, as well as specialist CAD software and the animation tool Blender.

INTERESTS

Outdoors: Keen rock climber, hiker, with a fascination for the natural world especially birds and fungi.

Music: Recreational piano and guitar.

PUBLICATIONS

Oral Presentations

"High Frequency Solutions to Electron Beam Position Measurements in the AWAKE Common Beamline", IOP-PAB 2024.

First Author Conference papers (dual- formal posters)

Spear, B. et al, "Beam studies using a Cherenkov diffraction based beam position monitor for AWAKE", in Proc. IPAC'24, Nashville, TN, May 2024, pp. 2327-2330. doi:10.18429/JACoW-IPAC2024-WEPG49.

Spear, B. et al, "Electron bunch position determination using a high frequency button beam position monitor in the AWAKE facility.", JACoW, vol. IBIC2024, Dec. 2024. <https://doi.org/10.18429/JACoW-IBIC2024-TUP21>.

Additionally, I have contributed to two Peer- Reviewed Papers

REFERENCES

Prof Philip N Burrows, DPhil Supervisor and Director of the John Adams Institute for Accelerator Science, Department of Physics, University of Oxford, Denys Wilkinson Building, Oxford Street, Oxford, UK

Email: philip.burrows@physics.ox.ac.uk

Manfred Wendt, DPhil Supervisor (CERN-based) Current EIC Instrumentation Systems Engineer at Brookhaven National Laboratory, NY, USA.

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